



CSEE4119: Computer Networks

Class Project

Xia Zhou
Department of Computer Science
Columbia University



Blockchain!

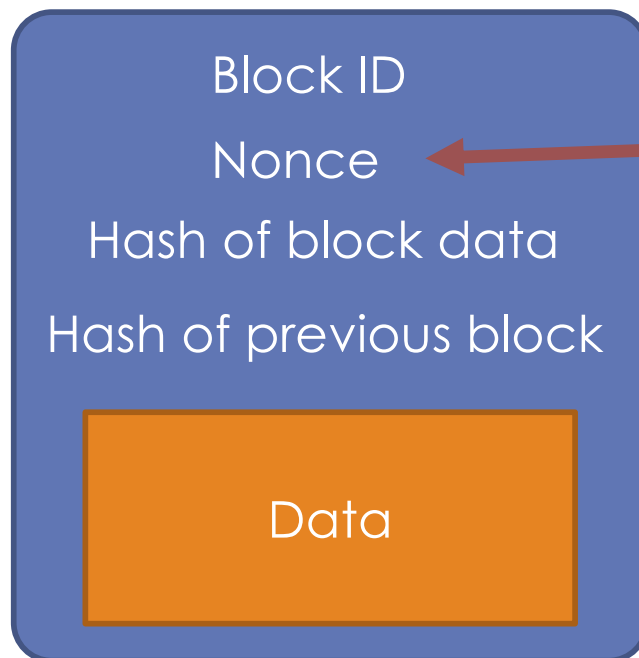
Implement a simplified blockchain and build a demo application of your choice

Blockchain

- A distributed mechanism that allows a network of nodes to agree on the state even if the nodes do not trust each other
- Key elements
 - Linked records using cryptography (hash)
 - Peer-to-peer

How does it Work?

- A linked list of records (blocks)



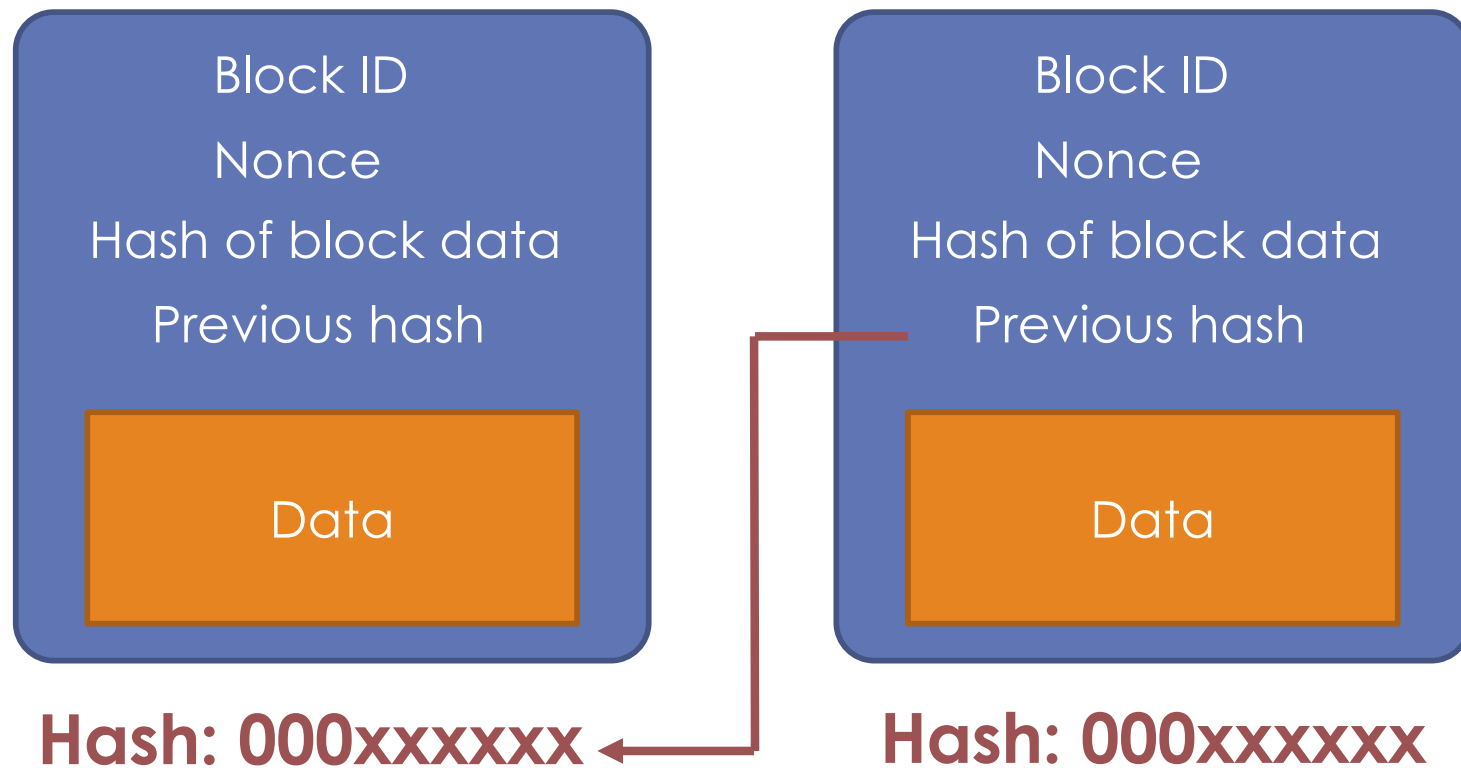
Nonce is selected so that the hash of the **block header** has a unique pattern (e.g., first x bits are zeros)

The process of finding a valid nonce is called **mining**

Hash: 000xxxxxx

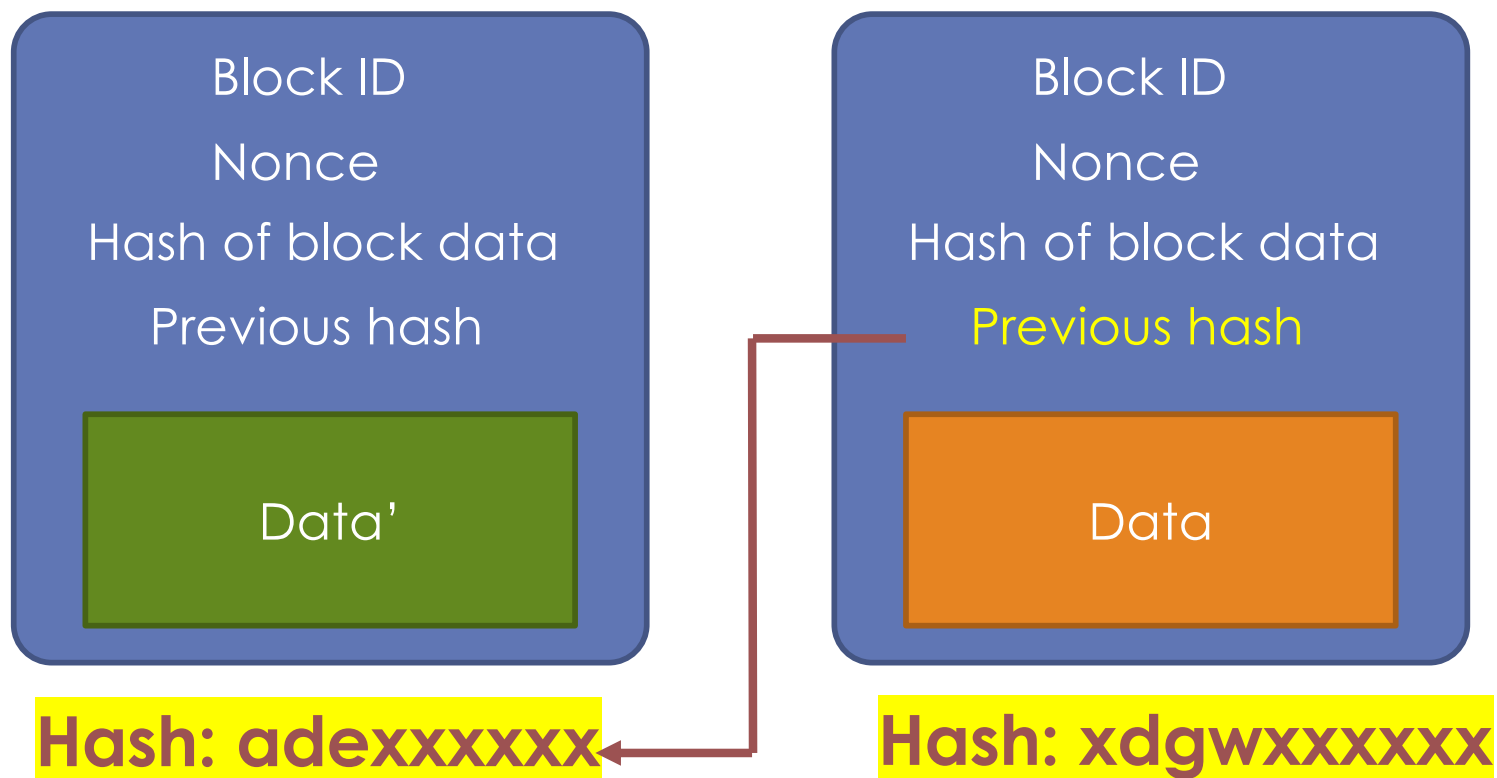
How does it Work?

- A linked list of records (blocks)
- Each block contains hash of the previous block



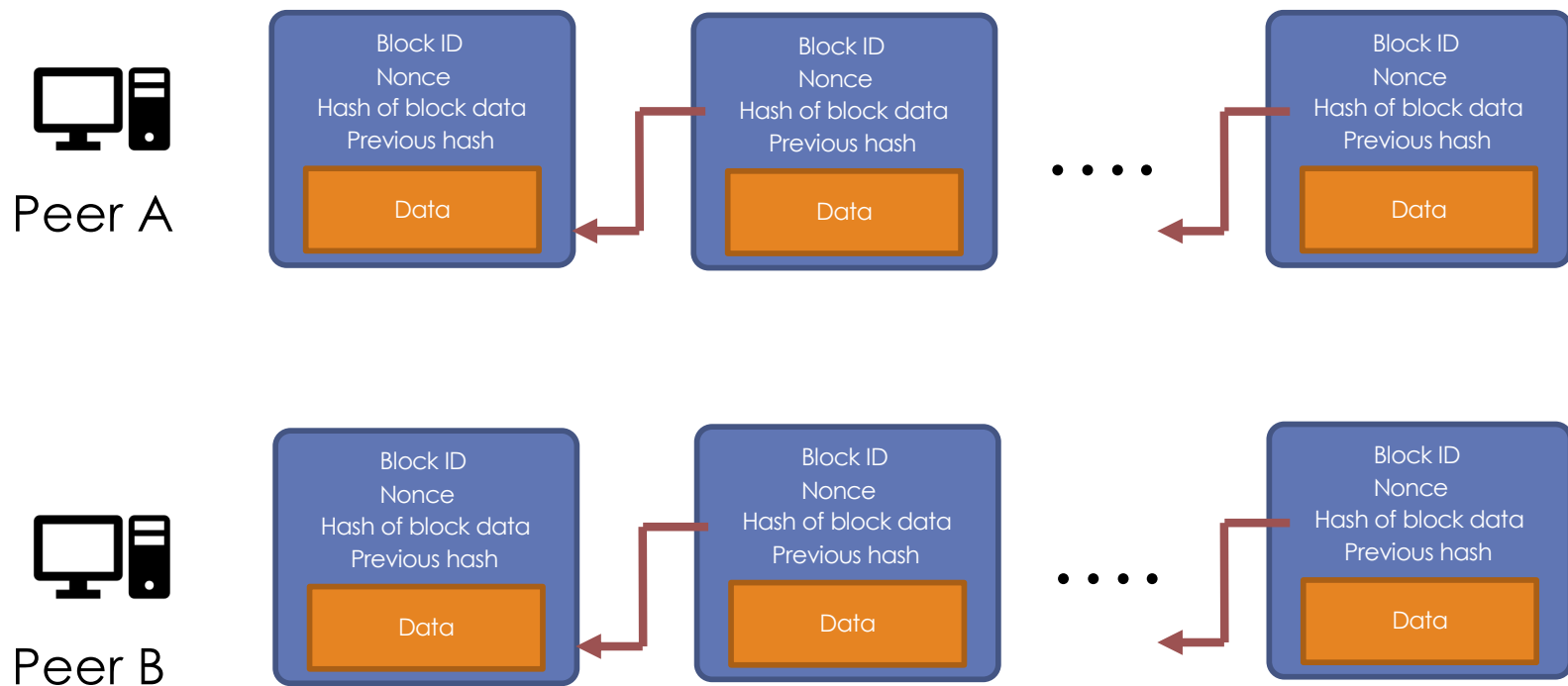
How does it Work?

- A linked list of records (blocks)
- Change in one block affects all subsequent blocks!



Peer-to-Peer Blockchain

- Blockchain managed by a P2P network, forming a **distributed ledger**
- Every node stores its copy of the blockchain



Peer-to-Peer Blockchain

- Blockchain managed by a P2P network, forming a **distributed ledger**
- Every node stores its copy of the blockchain
- Each node can add a block to the chain and then broadcast it to all other nodes
- Adding a block requires **proof-of-work (mining)**
- Other nodes verify the block before adding it to their local blockchains

Project Requirements

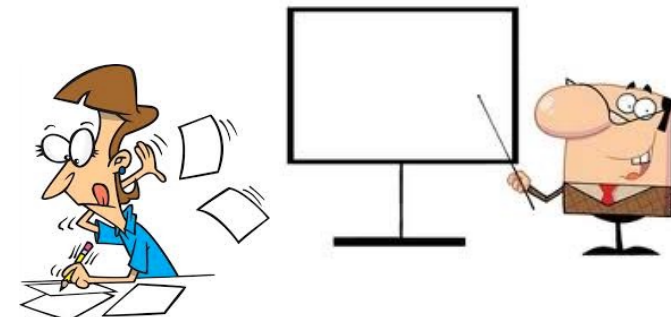
- **Basic** demonstration of the P2P blockchain technology
 - Implementing a P2P network with 1 tracker and at least 3 peers
 - Implementing a basic blockchain (creation, verification, broadcast to other peers, etc)
 - A demo application based on your blockchain protocol
 - e.g., cryptocurrency, voting, ...
 - Demonstrate how the use of blockchain is resilient to invalid modifications and transactions

Bonus Features

- Graphical application interface
- Dynamic adjustment of mining difficulty based on the available computation power
- Including multiple transactions inside a block and verify a transaction using Merkle tree
- Any other features that you can think of...

Plan for Project Weeks

- Each team is assigned with a TA shepherd
- Each team creates a **private GitHub repo** and share it with the shepherd
- By April 21: meet with your shepherd to go over design ideas and work partition
- By April 23: commit your DESIGN.md to GitHub
- May 4: due date for project source code and documentation and project demo
- By May 7: Fill in a team evaluation form



Project Demo and Presentation

- Each team has 10 minutes with the shepherd
 - **Live demo** of your blockchain application
 - A slide deck to describe your project design, implementation, testing, work partition, and lessons learned
 - Q/A: shepherd can ask questions about your code implementation, overall design, test strategies
- Important part of the project grade

Have fun while learning
something new!

Effective
teamwork is key

